

- 3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.
 (b) Use these equations to find v_x and i_x .

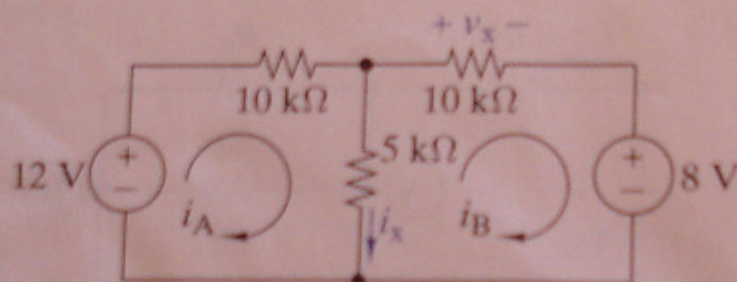


FIGURE P3-9

- 3-16 (a) Formulate mesh-current equations for the circuit in Figure P3-16.
 (b) Use these equations to find the input resistance.

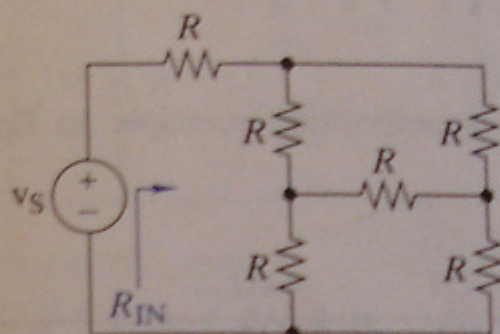


FIGURE P3-16

3-17 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

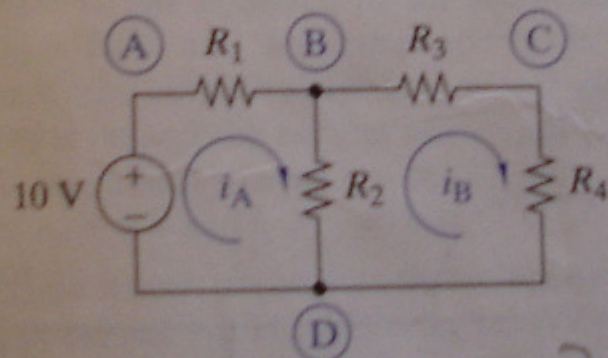


FIGURE P3-17

3-20 Find the mesh currents i_A , i_B , and i_C in Figure P3-20.

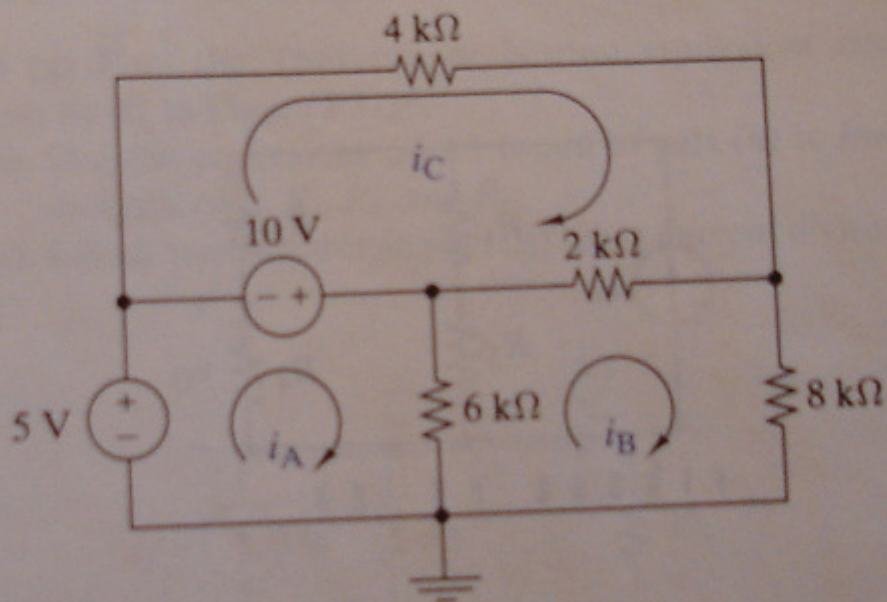


FIGURE P3-20

3-27 Use the superposition principle to find i_O in Figure P3-27.

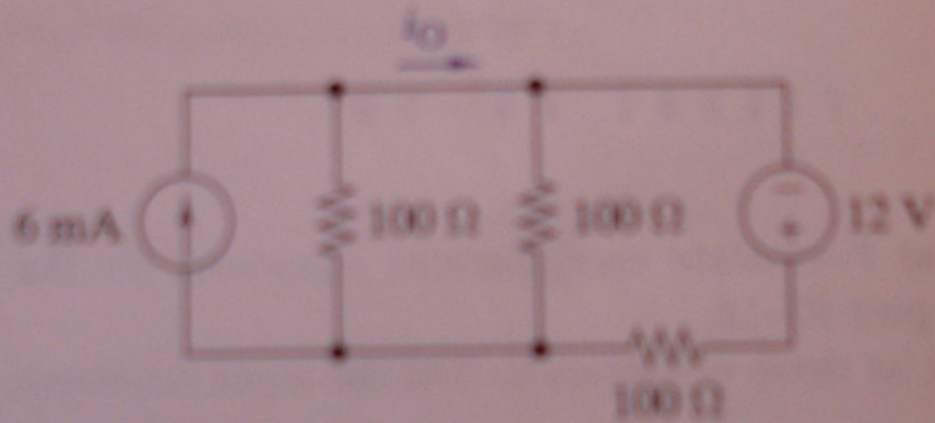


FIGURE P3-27

3-30 Use the superposition principle to find v_O in terms of v_1 , v_2 , and R in Figure P3-30.

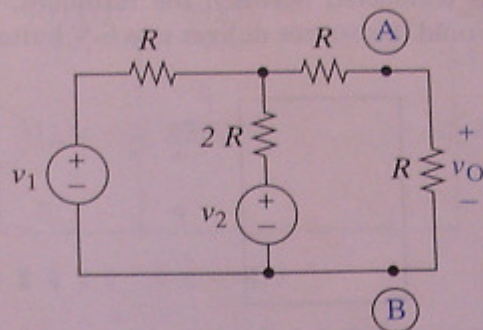


FIGURE P3-30

3-33 A linear circuit is driven by an independent voltage source $v_s = 10$ V and an independent current source $i_s = 10$ mA. The output voltage is $v_O = 2$ V when the voltage source is on and the current source off. The output is $v_O = 1$ V when both sources are on. Find the output voltage when $v_s = 5$ V and $i_s = -10$ mA.

4-2 Find the voltage gain v_O/v_1 and the current gain i_O/i_1 in Figure P4-2. For $i_1 = 2 \text{ mA}$, find the power supplied by the input current source and the power delivered to the $2\text{-k}\Omega$ load resistor.

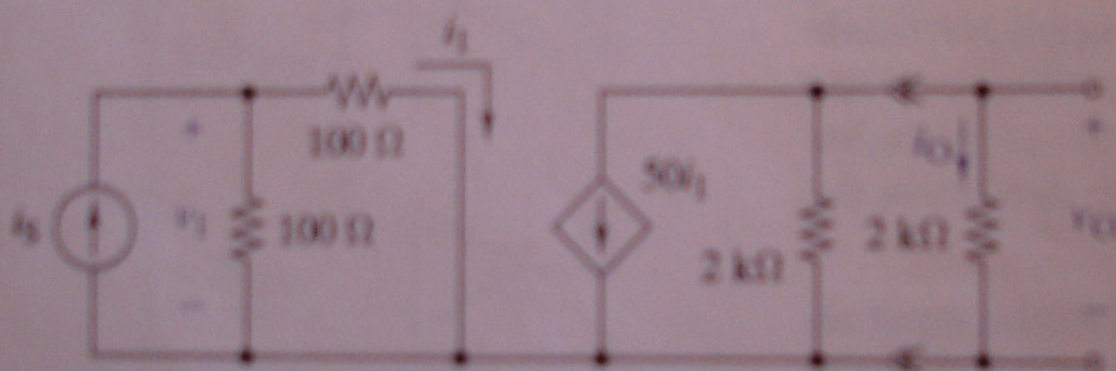


FIGURE P4-2

4-8 Find an expression for the current gain i_O/i_S in Figure P4-8.

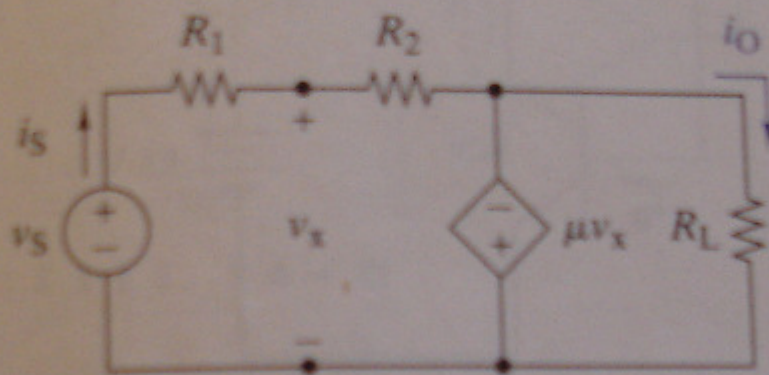


FIGURE P4-8

4-9 Find an expression for the voltage gain v_O/v_S in Figure P4-9.

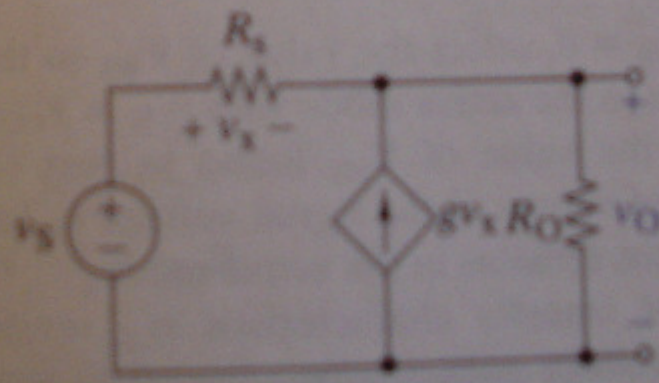


FIGURE P4-9